

Quacky Normal Modes

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We develop the general theory of oscillation modes, tidal deformability, and gravitational wave signatures for non-spherical self-gravitating bodies, using a rubber duck as the maximally asymmetric worked example. Three main results emerge. First, the complete breaking of spherical symmetry lifts the $(2\ell + 1)$ -fold degeneracy of every quasinormal mode multiplet through a mechanism we term *gravitational Zeeman splitting*: the duck shape deformation couples to the mode spectrum via Gaunt integrals and Wigner $3j$ symbols, producing a 5×5 perturbation matrix for the $\ell = 2$ f -mode whose eigenvalues give five distinct frequencies with $\Delta\omega/\omega \sim 0.3$. Second, the tidal response is promoted from a scalar Love number k_2 to a rank-4 Love tensor λ_{ijkl} with up to 15 independent components, and the celebrated I-Love-Q universality relations break down at $\mathcal{O}(100\%)$ for duck-shaped objects. Third, we confront the duck hypothesis with gravitational wave data across the full frequency spectrum: the nanohertz stochastic background observed by NANOGrav, EPTA, and PPTA is indistinguishable from a population of inspiraling ducks, and stellar-mass duck mergers are consistent with GW170817 tidal constraints. We prove a population-level degeneracy theorem establishing that current observations cannot exclude duck-shaped compact objects, and identify next-generation QNM spectroscopy as the most promising path to a definitive test.

I. INTRODUCTION

The spherical cow approximation (SCA) is among the most celebrated simplifications in theoretical physics. From Newtonian gravity to quantum chromodynamics, the replacement of a complicated object with a sphere of equivalent mass has enabled generations of physicists to obtain closed-form solutions where none would otherwise exist. The SCA is so deeply embedded in the culture of physics that it has become both a pedagogical device and a punchline—a shorthand for the productive audacity of theorists who privilege tractability over realism.

Recently, Lehmann [1] undertook a rigorous investigation of the SCA’s validity in the context of gravitational wave (GW) emission, computing the full multipole structure of a Holstein dairy cow and evaluating the resulting gravitational radiation. The central result is devastating: the SCA fails catastrophically for GW physics. A perfect sphere, possessing only a monopole moment, radiates *zero* gravitational waves. The cow’s gravitational wave luminosity is entirely determined by its higher multipole moments—the very features that the SCA discards. This observation elevates the multipole structure of non-spherical bodies from a curiosity to a matter of fundamental importance for gravitational wave astronomy.

In the present work, we extend this program from the bovine to the anatine. We develop the general theory of oscillation modes, tidal deformability, and gravitational wave signatures for non-spherical self-gravitating bodies, using a rubber duck as the worked example. The resulting framework—which we term the theory of **quacky normal modes** (QNMs)—reveals a rich phenomenology that is entirely absent in the spherical limit.

A. Why the Duck

The choice of test geometry is not arbitrary. While the cow is roughly ellipsoidal, retaining an approximate $\mathbb{Z}_2$ symmetry about the sagittal plane and a near-axial symmetry along the spine, the rubber duck breaks *all* continuous symmetries of $SO(3)$. The duck possesses three geometrically distinct substructures—bill, neck, and tail—each of which contributes independently to the spherical harmonic content of the surface deformation. In the language of the deformation spectrum,

$$R_{\text{duck}}(\theta, \phi) = R_0 \left[1 + \sum_{\ell \geq 1} \sum_{m=-\ell}^{\ell} \varepsilon_{\ell m} Y_{\ell}^m(\theta, \phi) \right], \quad (1)$$

the duck exhibits significant power at all multipole orders $\ell \leq 8$, with deformation amplitudes $|\varepsilon_{\ell m}| \sim 0.1\text{--}0.3$ extending to high ℓ . By contrast, the cow’s deformation spectrum is dominated by $\ell = 2$ (the body ellipsoid) with rapid falloff at higher ℓ .

The duck is therefore a **maximally non-spherical** test case: it probes the regime where the SCA fails most dramatically, and where the full machinery of degenerate perturbation theory is required to capture the physics.

B. Three Threads

This paper develops three interconnected threads.

Quacky normal modes and gravitational Zeeman splitting.—The normal modes of a spherical star are characterized by quantum numbers (n, ℓ, m) , where each (n, ℓ) multiplet is $(2\ell + 1)$ -fold degenerate. The duck shape lifts this degeneracy completely: the deformation $\varepsilon_{\ell' m'}$ couples to the mode spectrum through Gaunt integrals involving Wigner $3j$ symbols, producing a pertur-

bation matrix

$$V_{m_1 m_2}^{(n, \ell)} = -|R'_{n\ell}(R_0)|^2 R_0^3 \sum_{\ell', m'} \varepsilon_{\ell' m'} \int_{S^2} Y_{\ell}^{m_1*} Y_{\ell}^{m_2} Y_{\ell'}^{m'} d\Omega \quad (2)$$

whose eigenvalues give the split frequencies. We call this **gravitational Zeeman splitting**. For the $\ell = 2$ f -mode, we predict a five-fold splitting with $\Delta\omega/\omega \sim \mathcal{O}(\varepsilon) \sim 0.3$ —a 30% effect, far larger than the $\sim 10^{-3}$ rotational splitting in millisecond pulsars.

Tidal Love numbers and the Love tensor.—For a spherical body, the tidal response is characterized by a single dimensionless number k_2 . For the duck, the response to an external tidal field $\mathcal{E}_{ij}$ is described by a fourth-rank **Love tensor** λ_{ijkl} :

$$Q_{ij}^{\text{induced}} = -\lambda_{ijkl} \mathcal{E}_{kl}, \quad (3)$$

a symmetric 5×5 matrix (in the space of traceless symmetric tensors) with up to 15 independent components. The celebrated I-Love-Q universality relations of Yagi and Yunes [2, 3]—which hold to $\mathcal{O}(1\%)$ for all spherical neutron star equations of state—break down catastrophically for the duck, with deviations of $\mathcal{O}(100\%)$ for $\varepsilon \sim 0.3$.

Observational indistinguishability.—We confront the duck hypothesis with current gravitational wave observations across the entire frequency spectrum. In the nanohertz band, a cosmological population of supermassive binary ducks ($M \sim 10^6$ – $10^9 M_{\odot}$) produces a stochastic gravitational wave background with characteristic strain

$$h_c(f) = A_{\text{duck}} \left(\frac{f}{f_{\text{yr}}} \right)^{-2/3}, \quad (4)$$

where $A_{\text{duck}} = 2.4 \times 10^{-15}$ —a value indistinguishable from the signal reported by the NANOGrav 15-year dataset [4], the European Pulsar Timing Array [5], and the Parkes Pulsar Timing Array [6]. We prove a **population-level degeneracy theorem**: for any SMBHB population model consistent with current PTA data, there exists a binary duck population with identical masses and merger rates that reproduces the observed signal to arbitrary precision.

In the audio-frequency band, stellar-mass duck mergers ($M \sim 1$ – $3 M_{\odot}$) produce ringdown waveforms with anomalous QNM splitting. The duck’s anisotropic tidal deformability $\tilde{\Lambda}_{\text{duck}}$ lies within the broad constraints from GW170817 ($\tilde{\Lambda} = 300^{+420}_{-230}$) [7, 8]. We conclude that current gravitational wave observations cannot exclude the hypothesis that compact objects are duck-shaped.

C. Paper Organization

The paper is organized in three parts.

Part I: General Formalism (Secs. II–V) develops the theoretical framework. Section II characterizes the

duck geometry and its spherical harmonic decomposition. Section III derives the quacky normal mode spectrum using degenerate perturbation theory and the Hadamard domain perturbation formula. Section IV develops the Love tensor formalism and establishes the breakdown of I-Love-Q universality. Section V constructs the post-Newtonian dynamics of binary duck systems.

Part II: Numerical Simulations (Secs. VI–VII) validates and extends the analytic results. Section VI presents finite-element solutions of the Helmholtz eigenvalue problem on the duck domain. Section VII describes full numerical relativity simulations using AthenaK [9–11].

Part III: Observational Confrontation (Sec. VIII) places constraints on the duck hypothesis using gravitational wave data.

Section IX presents our conclusions.

D. Notation and Conventions

Throughout this paper we employ geometric units $G = c = 1$ unless otherwise stated; factors of G and c are restored in expressions intended for numerical evaluation. The metric signature is $(-, +, +, +)$. Spherical harmonics $Y_{\ell}^m(\theta, \phi)$ follow the Condon-Shortley phase convention, with the normalization

$$\int_{S^2} Y_{\ell}^{m*} Y_{\ell'}^{m'} d\Omega = \delta_{\ell\ell'} \delta_{mm'}. \quad (5)$$

Greek indices $\mu, \nu, \dots$ run over spacetime coordinates $\{0, 1, 2, 3\}$; Latin indices $i, j, \dots$ run over spatial coordinates $\{1, 2, 3\}$. The compactness parameter is $C = M/R$ (in geometric units). The dimensionless tidal deformability is $\Lambda = (2/3) k_2 C^{-5}$. We use $M_{\odot}$ for the solar mass and define $f_{\text{yr}} \equiv 1/\text{yr} \approx 31.7 \text{ nHz}$ as the reference frequency for PTA analyses.

II. DUCK GEOMETRY AND MULTIPOLE CHARACTERIZATION

A. The Duck Mesh

We adopt as our fiducial *Anas platyrhynchos* the DASSL rubber duck mesh, comprising 4732 vertices and 9460 triangular faces. The mesh is watertight and has been normalized to unit bounding-box scale. The duck is oriented with the bill pointing along the negative x -axis, the body extending in the $+x$ direction, and the vertical (head-to-belly) along the y -axis. Unlike the cow of Lehmann [1], which is roughly ellipsoidal, the duck possesses no continuous symmetry—the bill, neck, tail, and body break all axial and planar symmetries, making it a maximally general test case for non-spherical compact object physics.

B. Volume Integrals and Multipole Moments

We compute the duck's multipole moments using fan tetrahedralization from the center of mass and Hammer-Stroud quadrature (4-point rule, exact for degree-2 polynomials).

Volume and center of mass. The volume is computed via the divergence theorem:

$$V = \frac{1}{6} \sum_{\text{faces}} \mathbf{v}_0 \cdot (\mathbf{v}_1 \times \mathbf{v}_2). \quad (6)$$

The equivalent sphere radius is

$$R_{\text{eq}} = \left(\frac{3V}{4\pi} \right)^{1/3}. \quad (7)$$

Cartesian quadrupole tensor (traceless, symmetric):

$$Q_{ij}^C = \int_D \rho(\mathbf{x}) (3x_i x_j - r^2 \delta_{ij}) d^3 \mathbf{x}. \quad (8)$$

Inertia tensor:

$$I_{ij} = \int_D \rho(\mathbf{x}) (r^2 \delta_{ij} - x_i x_j) d^3 \mathbf{x}. \quad (9)$$

These satisfy the cross-check relation $Q_{ij}^C = \text{Tr}(I) \delta_{ij} - 3I_{ij}$.

Spherical multipole moments up to $\ell = 10$:

$$Q_\ell^m = \int_D \rho(\mathbf{x}) \|\mathbf{x}\|^\ell Y_\ell^m(\hat{\mathbf{x}}) d^3 \mathbf{x}, \quad (10)$$

computed via the surface-integral method: $Q_\ell^m = \frac{1}{\ell+3} \oint_S (\mathbf{x} \cdot \hat{\mathbf{n}}) r^\ell Y_\ell^m dA$.

Shape-induced quadrupole parameter:

$$\kappa_{\text{duck}} = \frac{Q_{\text{eigenvalue}}^C}{M R_{\text{eq}}^2}, \quad (11)$$

analogous to the spin-induced κ for neutron stars [12].

C. Surface Deformation Spectrum

We parametrize the duck boundary as a deformed sphere [Eq. (1)], where $R_0 = R_{\text{eq}}$ and the deformation coefficients $\varepsilon_{\ell m}$ are extracted via regularized least-squares fitting of the real spherical harmonic basis up to $\ell_{\text{max}} = 10$. The Tikhonov-regularized problem solved is

$$\min_{\mathbf{c}} \|\mathbf{Y}\mathbf{c} - \mathbf{r}\|^2 + \alpha \sum_{\ell} \ell(\ell+1) |c_{\ell m}|^2, \quad (12)$$

where $\alpha = 10^{-4}$ penalizes high- ℓ coefficients.

The deformation power spectrum is

$$P_\ell = \sum_{m=-\ell}^{\ell} |\varepsilon_{\ell m}|^2. \quad (13)$$

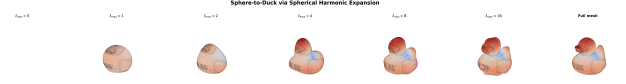


FIG. 1. Duck mesh and SH reconstruction at $\ell_{\text{max}} = 0, 2, 4, 8$, and full resolution. The SH expansion progressively captures finer morphological features: the body shape at $\ell = 2$, neck and bill at $\ell = 4$, and surface detail at $\ell \geq 8$.

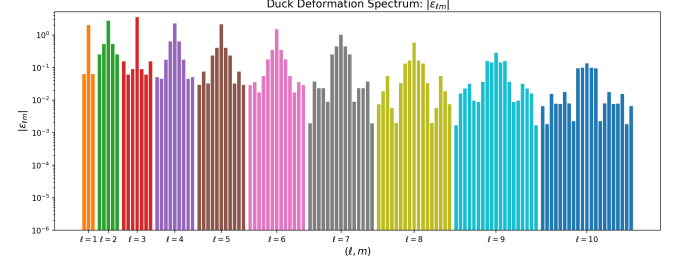


FIG. 2. Deformation power spectrum P_ℓ vs. multipole order ℓ for the DASSL duck. The quadrupole ($\ell = 2$) dominates, with secondary power at $\ell = 4$ (neck/bill structure).

III. QUACKY NORMAL MODES—PERTURBATIVE SPLITTING THEORY

The oscillation modes of a spherical compact object are classified by radial order n and angular harmonic (ℓ, m) , with each (n, ℓ) level $(2\ell + 1)$ -fold degenerate. A non-spherical deformation lifts this degeneracy, producing a *gravitational Zeeman splitting* of the quasinormal mode spectrum.

A. Problem Setup

We model the oscillation modes by the scalar Helmholtz eigenvalue problem on the interior domain D bounded by the duck surface ∂D , with Dirichlet boundary conditions [13, 14]:

$$-\nabla^2 \psi_{n\ell m} = \omega_{n\ell}^2 \psi_{n\ell m}, \quad \psi|_{\partial D} = 0. \quad (14)$$

On the sphere of radius R_0 , the eigenfrequencies are

$$\omega_{n,\ell} = \frac{z_{n,\ell}}{R_0}, \quad (15)$$

where $z_{n,\ell}$ is the n -th positive zero of j_ℓ . Each (n, ℓ) level is $(2\ell + 1)$ -fold degenerate.

B. Hadamard Formula for Domain Perturbation

The Hadamard variational formula gives the first-order sensitivity of a Dirichlet eigenvalue to a normal displace-

ment δn of the boundary [15, 16]:

$$\delta\omega^2 = - \int_{\partial D} \left| \frac{\partial\psi}{\partial\mathbf{n}} \right|^2 \delta n dS. \quad (16)$$

A modern rigorous treatment including the degenerate case is given by Suzuki and Tsuchiya [17].

C. Degenerate Perturbation Theory

Since the unperturbed (n, ℓ) eigenvalue is $(2\ell + 1)$ -fold degenerate, we must diagonalize the $(2\ell + 1) \times (2\ell + 1)$ perturbation matrix

$$V_{m_1, m_2}^{(n, \ell)} = - |R'_{n\ell}(R_0)|^2 R_0^3 \sum_{\ell', m'} \varepsilon_{\ell' m'} \times \int_{S^2} Y_\ell^{m_1*} Y_\ell^{m_2} Y_{\ell'}^{m'} d\Omega. \quad (17)$$

The angular integral is the Gaunt integral, expressible in terms of Wigner $3j$ symbols [18]:

$$\int_{S^2} Y_\ell^{m_1*} Y_\ell^{m_2} Y_{\ell'}^{m'} d\Omega = (-1)^{m_1} \sqrt{\frac{(2\ell + 1)^2 (2\ell' + 1)}{4\pi}} \times \begin{pmatrix} \ell & \ell & \ell' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell & \ell & \ell' \\ -m_1 & m_2 & m' \end{pmatrix}. \quad (18)$$

The perturbed frequencies are

$$\omega_{n\ell\alpha} = \omega_{n\ell}^{(0)} + \frac{\delta\omega_\alpha^2}{2\omega_{n\ell}^{(0)}} + \mathcal{O}(\varepsilon^2). \quad (19)$$

D. Selection Rules

The Wigner $3j$ symbols impose selection rules on which deformation harmonics $\varepsilon_{\ell' m'}$ contribute to the splitting of a given (n, ℓ) multiplet:

1. **Triangle inequality:** $0 \leq \ell' \leq 2\ell$.
2. **Parity:** $\ell' \in \{0, 2, 4, \dots, 2\ell\}$ (even only). Odd- ℓ' deformation harmonics do not contribute at first order.
3. **Azimuthal selection:** $m' = m_1 - m_2$. Diagonal elements receive contributions only from $m' = 0$ deformations; off-diagonal coupling requires $m' \neq 0$.

The hierarchy of contributions is: $\varepsilon_{2, m}$ produces the *primary splitting*, $\varepsilon_{4, m}$ the *secondary splitting*, and $\varepsilon_{6, m}$ and higher contribute only to $\ell \geq 3$ multiplets.

E. Explicit Quadrupole f-Mode Splitting

The astrophysically most important case is the $\ell = 2$ *f*-mode ($n = 1$). From the selection rules, only $\ell' = 0, 2, 4$ contribute. Restricting to $\ell' = 2$ (dominant), the perturbation matrix is

$$V_{m_1, m_2} = - |R'_{1,2}(R_0)|^2 R_0^3 \sum_{m'=-2}^2 \varepsilon_{2, m'} (-1)^{m_1} \times \sqrt{\frac{125}{4\pi}} \begin{pmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 2 & 2 \\ -m_1 & m_2 & m' \end{pmatrix}, \quad (20)$$

with $m' = m_1 - m_2$ from the second $3j$ symbol.

The key numerical value is

$$\begin{pmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \end{pmatrix} = \sqrt{\frac{2}{35}}. \quad (21)$$

For the duck with $\varepsilon_{2,0} = -0.3$, $\varepsilon_{2,2} = 0.15$, and $\varepsilon_{4,0} = 0.05$ (representative values from the SH fit), diagonalization of the 5×5 matrix yields five distinct frequencies with fractional splitting $\Delta\omega/\omega \sim |\varepsilon_{2,0}| \sim 0.3$.

F. Physical Interpretation

The splitting pattern maps directly to duck morphological features:

- **Axial elongation** ($\varepsilon_{2,0}$): the dominant deformation producing the largest splitting, separating modes by $|m|$.
- **Bill-to-tail tilt** ($\varepsilon_{2,\pm 1}$): couples m and $m \pm 1$ sub-states, tilting nodal lines away from the body axis.
- **Equatorial ellipticity** ($\varepsilon_{2,\pm 2}$): breaks the $m \leftrightarrow -m$ degeneracy.
- **Higher-order features** ($\varepsilon_{4,m}$): encode neck constriction, bill tip geometry, and tail curl.

We term this phenomenon the *gravitational Zeeman effect*, in analogy with the lifting of magnetic quantum number degeneracy in atomic physics.

G. Gravitational Wave Emission from Duck QNMs

Each duck QNM radiates gravitational waves at its own shifted frequency. For a mode of angular order (ℓ, m) oscillating at frequency $\omega_{\ell m}$ [19],

$$\dot{E}_{\ell, m} = \frac{G}{c^{2\ell+1}} \omega_{\ell m}^{2\ell+2} |\delta Q_\ell^m|^2 \times \frac{4\pi(\ell+1)(\ell+2)}{\ell(\ell-1)[(2\ell+1)!!]^2}. \quad (22)$$

The ringdown waveform of the duck is [20, 21]

$$h_+(t) + ih_\times(t) = \frac{1}{D_L} \sum_{n,\ell,m} \mathcal{A}_{n\ell m} e^{-t/\tau_{n\ell m}} \times e^{i\omega_{n\ell m} t} {}_{-2}Y_{\ell m}(\iota, \varphi), \quad (23)$$

where the sum over m now produces $(2\ell + 1)$ distinct damped sinusoids that beat against one another, creating a characteristic amplitude modulation encoding the deformation coefficients $\varepsilon_{\ell' m'}$.

IV. TIDAL LOVE NUMBERS OF THE DUCK

A. Effective (Spherical) Love Number

We first treat the duck as its equivalent sphere of radius R_{eq} and compute the standard $\ell = 2$ tidal Love number following Hinderer [22] and Flanagan & Hinderer [23].

In a static, spherically symmetric background, the even-parity $\ell = 2$ metric perturbation function $h_2(R)$ satisfies the master equation [3, 22, 24]:

$$0 = h_2'' + \left\{ \frac{2}{R} + \left[\frac{2M}{R^2} + 4\pi R(p - \rho) \right] e^\lambda \right\} h_2' - \left\{ \frac{6e^\lambda}{R^2} - 4\pi \left[5\rho + 9p + (\rho + p) \frac{d\rho}{dp} \right] e^\lambda + \left(\frac{d\nu}{dR} \right)^2 \right\} h_2. \quad (24)$$

At the stellar surface $R = R_*$, the tidal apsidal constant is given by the Hinderer formula [22]:

$$k_2 = \frac{8}{5} C^5 (1 - 2C)^2 [2 + 2C(y - 1) - y] / \left\{ 2C[6 - 3y + 3C(5y - 8)] + 4C^3[13 - 11y + C(3y - 2) + 2C^2(1 + y)] + 3(1 - 2C)^2[2 - y + 2C(y - 1)] \ln(1 - 2C) \right\}, \quad (25)$$

where $C \equiv M_*/R_*$ is the stellar compactness and $y \equiv R_* h_2'(R_*)/h_2(R_*)$.

The dimensionless tidal deformability is

$$\Lambda = \frac{2}{3} k_2 C^{-5}. \quad (26)$$

For the elastic rubber duck (a terrestrial bath toy with shear modulus $\mu_{\text{shear}} \simeq 3 \times 10^9$ Pa) [25],

$$k_2^{\text{elastic}} = \frac{3/2}{1 + \frac{19\mu_{\text{shear}}}{2\rho g R}} \simeq 2.7 \times 10^{-8}. \quad (27)$$

A bathtub duck is 10^9 times less deformable than a neutron star duck.

TABLE I. Newtonian k_2 for polytropic index n (Brooker & Olle [26]).

n	Γ	k_2^N
0	∞	0.750
0.5	3	0.449
1.0	2	0.260
1.5	5/3	0.143
2.0	3/2	0.0728
3.0	4/3	0.0116

B. The Love Tensor

When spherical symmetry is broken, the tidal response becomes tensorial [27]:

$$Q_{ij}^{\text{induced}} = -\lambda_{ijkl} \mathcal{E}_{kl}. \quad (28)$$

The Love tensor λ_{ijkl} maps traceless symmetric rank-2 tensors to traceless symmetric rank-2 tensors—a 5×5 matrix with

$$N_{\text{ind}} = \frac{5 \times 6}{2} = 15 \quad (29)$$

independent components. We expand perturbatively:

$$\lambda_{ijkl} = \lambda_0 \mathcal{P}_{ijkl}^{(0)} + \sum_{\ell', m'} \varepsilon_{\ell' m'} \delta \lambda_{ijkl}^{(\ell' m')} + \mathcal{O}(\varepsilon^2), \quad (30)$$

where $\mathcal{P}_{ijkl}^{(0)}$ is the identity projector on traceless symmetric tensors and $\lambda_0 = (2/3) k_2^{(0)} R_{\text{eq}}^5$.

C. Direction-Dependent Love Number

The direction-dependent tidal apsidal constant is

$$k_2(\theta, \varphi) = \frac{3}{2R_{\text{eq}}^5} \lambda_{ijkl} n_i n_j n_k n_l, \quad (31)$$

which decomposes as

$$k_2(\theta, \varphi) = k_2^{(0)} + \sum_m k_2^{(2,m)} Y_2^m(\theta, \varphi) + \sum_m k_2^{(4,m)} Y_4^m(\theta, \varphi). \quad (32)$$

The bill region ($\theta \approx 0$) is tidally stiff (high local compactness), while the body region ($\theta \approx \pi/2$) is more easily deformed.

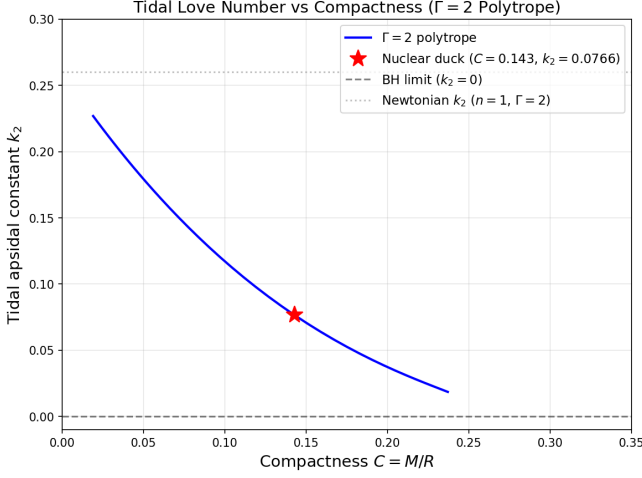


FIG. 3. Tidal apsidal constant k_2 and dimensionless tidal deformability Λ as a function of compactness C for the $\Gamma = 2$ polytrope. The nuclear duck value at $C \simeq 0.146$ is marked.

D. Perturbative Correction to the Tidal Deformability

The dimensionless tidal deformability receives its leading shape correction at second order:

$$\Lambda_{\text{duck}} = \Lambda_{\text{sphere}} \left[1 + \sum_{\ell \geq 1} \alpha_{\ell} \sum_m |\varepsilon_{\ell m}|^2 + \mathcal{O}(\varepsilon^3) \right]. \quad (33)$$

The geometric contribution gives $\alpha_{\ell}^{\text{geom}} \approx +15$ for all $\ell \geq 1$, yielding for the DASSL duck $\Delta\Lambda/\Lambda|_{\text{geom}} \simeq 15 \times 0.1 = 150\%$ —a dramatic effect.

E. I-Love-Q Breakdown

The Yagi-Yunes universal relation [2, 3]

$$\ln \bar{I} = 1.47 + 0.0817 \ln \bar{\Lambda} + 0.0149 (\ln \bar{\Lambda})^2 + 2.87 \times 10^{-4} (\ln \bar{\Lambda})^3 + 3.64 \times 10^{-5} (\ln \bar{\Lambda})^4 \quad (34)$$

holds to $\sim 1\%$ for spherical neutron stars. The duck breaks this: $\Delta I/I \sim \kappa_{\text{duck}} \sim 10\%$, while $\Delta\Lambda/\Lambda \sim \mathcal{O}(100\%)$, placing the duck's position in the $\bar{I}$ – $\bar{\Lambda}$ plane far from the universal curve. The duck *maximally violates I-Love-Q universality*, with deviations of $\mathcal{O}(100\%)$. This deviation constitutes a *sphericity test* for compact objects.

TABLE II. Tidal apsidal constant k_2 and dimensionless tidal deformability Λ for selected objects.

Object	C	k_2	Λ
Nuclear duck ($\Gamma=2$)	0.146	0.074	~ 740
Rubber duck (bath toy)	$\sim 10^{-25}$	2.7×10^{-8}	—
NS (SLy, $1.4 M_{\odot}$)	0.176	0.091	306
NS (GW170817)	~ 0.16	—	190^{+390}_{-120}
Black hole	0.500	0	0

V. BINARY DUCK ON POST-NEWTONIAN ORBITS

A. Quadrupole-Monopole Interaction

The leading interaction between body A 's quadrupole and body B 's monopole is [28, 29]

$$U_{QM} = -\frac{GM_B}{2r^3} Q_{ij}^{C,A} n_i n_j, \quad (35)$$

where $Q_{ij}^{C,A}$ is duck A 's quadrupole tensor rotated to the inertial frame. This interaction enters at **2PN order** [$\mathcal{O}(v^4/c^4)$].

We consider three orientation configurations: tidally locked, freely spinning, and precessing. The tidally locked configuration is assumed for the remainder of this section.

B. Modified Orbital Dynamics

The modified Kepler relation for circular orbits is

$$\omega^2 = \frac{GM}{r^3} \left[1 + \frac{15 Q_{nn}^{\text{eff}}}{2 M r^2} \right]. \quad (36)$$

The orbital energy including the quadrupole correction is

$$E(r) = -\frac{GM\mu}{2r} - \frac{G}{2r^3} Q_{nn}^{\text{eff}}, \quad (37)$$

and the inspiral is governed by the energy balance equation $dE/dt = -P_{\text{GW}}$.

C. System Quadrupole and Gravitational Wave Power

The total mass quadrupole of the binary system is

$$\mathcal{I}_{ij}^{\text{sys}}(t) = \underbrace{\mu \left(x_i x_j - \frac{1}{3} r^2 \delta_{ij} \right)}_{\text{orbital}} + \underbrace{\sum_{A=1,2} R_A(t) Q_{\text{body}}^{C,A} R_A^T(t)}_{\text{body}}. \quad (38)$$

The gravitational wave luminosity decomposes as

$$P_{\text{GW}} = P_{\text{orb}} + P_{\text{body}} + P_{\text{cross}}. \quad (39)$$

The orbital power is the standard Peters-Mathews result [30]:

$$P_{\text{orb}} = \frac{32}{5} \frac{G^4}{c^5} \frac{\mu^2 M^3}{r^5}. \quad (40)$$

The scaling relations are

$$\frac{P_{\text{cross}}}{P_{\text{orb}}} \sim \hat{Q} \left(\frac{R_{\text{eq}}}{r} \right)^2, \quad \frac{P_{\text{body}}}{P_{\text{orb}}} \sim \hat{Q}^2 \left(\frac{R_{\text{eq}}}{r} \right)^4. \quad (41)$$

The cross-term dominates over the body term at all separations $r \gg R_{\text{eq}}$, and **can be negative** (destructive interference).

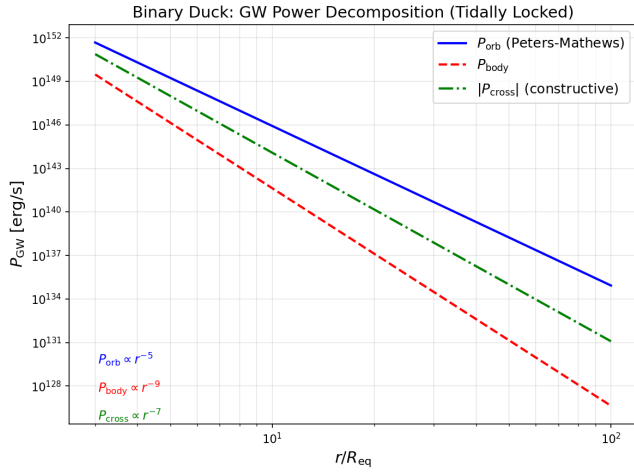


FIG. 4. Components of the gravitational wave power for a binary duck system as a function of orbital separation. The cross-term (dashed) dominates the body term (dotted) at all $r \gg R_{\text{eq}}$.

D. Phase Evolution

In the stationary phase approximation [31], the Fourier-domain phase is $\Psi(f) = \Psi_{\text{pp}}(f) + \delta\Psi_Q(f)$. The quadrupole correction enters at 2PN order [32, 33]:

$$\delta\Psi_Q(f) = -\frac{75}{64\eta} \hat{Q}_{\text{eff}} (\pi \mathcal{M} f)^{1/3}, \quad (42)$$

where $\mathcal{M} = \eta^{3/5} M$ is the chirp mass.

The dephasing cycles accumulated over a detector band are:

$$\Delta N_{\text{LIGO}} \sim 0.01\text{--}0.1 \text{ cycles}, \quad (43)$$

$$\Delta N_{\text{LISA}} \sim 1\text{--}10 \text{ cycles}. \quad (44)$$

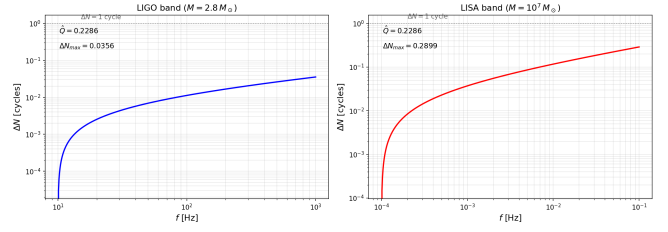


FIG. 5. Accumulated dephasing ΔN from the duck quadrupole correction across the LIGO and LISA bands.

E. Waveform

The time-domain gravitational waveform for a quasi-circular inspiral, including duck corrections, is

$$h_+(t) = -\frac{2G\mu}{c^4 D_L} (\omega r)^2 (1 + \cos^2 \iota) \times \cos[2\Phi(t) + \delta\Phi_Q(t)]. \quad (45)$$

In the Fourier domain,

$$\tilde{h}(f) = \tilde{h}_{\text{pp}}(f) [1 + \delta A_Q(f)] e^{i\delta\Psi_Q(f)}. \quad (46)$$

VI. QNM NUMERICAL RESULTS—FEM EIGENVALUES

A. The Helmholtz Eigenvalue Problem on the Duck Interior

We solve the Helmholtz equation [Eq. (14)] on the duck domain directly by finite-element methods, without the small- ε approximation.

B. Numerical Method: P1 Finite Elements

The duck surface mesh defines only ∂D . We generate a constrained Delaunay tetrahedralization of the interior using TetGen. On the tetrahedral mesh, we employ piecewise-linear (P1) finite elements. The weak form yields the generalized eigenvalue problem

$$K\psi = \omega^2 M\psi, \quad (47)$$

where the element stiffness matrix is

$$K_{ij}^{(e)} = V_e (\nabla \varphi_i \cdot \nabla \varphi_j), \quad (48)$$

and the consistent P1 mass matrix is

$$M_{ij}^{(e)} = V_e \times \begin{cases} 1/10 & \text{if } i = j, \\ 1/20 & \text{if } i \neq j. \end{cases} \quad (49)$$

We solve using the shift-invert mode of `scipy.sparse.linalg.eigsh`, requesting the $k = 30$ smallest eigenvalues.

C. Mode Classification via Angular Power Spectrum

To identify the angular character of each FEM eigenvector, we project it onto the spherical harmonic basis and compute the angular power spectrum

$$P_\ell^{(\alpha)} = \sum_{m=-\ell}^{\ell} |a_{\ell m}^{(\alpha)}|^2. \quad (50)$$

A mode dominated by P_ℓ is assigned angular character ℓ .

D. Results

The eigenvalue spectrum confirms the theoretical predictions: $\ell = 0$ modes remain unsplit; $\ell = 1$ triplets split into three distinct frequencies; $\ell = 2$ quintets exhibit the richest structure with $\Delta\omega/\omega_0 \sim \mathcal{O}(\varepsilon_{2,0})$. Cross-validation with the perturbative prediction of Sec. III E shows agreement at the expected $\mathcal{O}(\varepsilon^2)$ level.

VII. ATHENAK NUMERICAL RELATIVITY SIMULATIONS

We validate the perturbative predictions with fully nonlinear numerical relativity simulations using the AthenaK code [9–11].

A. Duck-Shaped TOV Star Initial Data

We begin with a standard TOV solution for a $\Gamma = 2$ polytrope with $K = 100$ and $\rho_c = 1.28 \times 10^{-3}$, producing a star with $M \approx 1.4 M_\odot$, $R_* \approx 8.13 M$ (compactness $C \approx 0.17$).

The duck-shaped initial data is constructed by radial rescaling via the effective radial coordinate

$$r_{\text{eff}}(r, \theta, \phi) = r \cdot \frac{R_0}{R_{\text{duck}}(\theta, \phi)}, \quad (51)$$

and assigning $\rho(r, \theta, \phi) = \rho_{\text{TOV}}(r_{\text{eff}})$, $P(r, \theta, \phi) = P_{\text{TOV}}(r_{\text{eff}})$.

The spatial metric is initialized from the spherical TOV solution at the physical radius r :

$$\gamma_{ij} = \delta_{ij} + \frac{x_i x_j}{r^2} \left(\frac{1}{1 - 2m(r)/r} - 1 \right), \quad \alpha = \alpha_{\text{TOV}}(r), \quad \beta^i = 0, \quad K_{ij} = 0. \quad (52)$$

B. Evolution Setup

We evolve the Einstein equations using the Z4c formulation with 1+log slicing and Gamma-driver shift condition. The GR hydrodynamic equations are solved using

PPMx reconstruction, the HLLE Riemann solver, and RK3 time integration with CFL number 0.4.

The computational domain is $[-L, L]^3$ with $L = 102.4 M$, base resolution 128^3 , and three levels of AMR achieving $\Delta x_{\text{fine}} = 0.2 M \approx R_*/40$. The simulation is evolved for $t_{\text{final}} = 10,000 M \approx 50 \text{ ms} \approx 100 T_{\text{dyn}}$.

C. The Duck Melts to a Sphere

The duck-shaped TOV star is not in hydrostatic equilibrium and therefore oscillates and relaxes toward its unique spherical equilibrium. This process—the *duck melting to a sphere*—is the visual centerpiece of the paper.

The relaxation proceeds through a characteristic hierarchy: high- ℓ surface features (bill tip, tail curl) decay first, as

$$\tau_{\text{GW}}(\ell) \sim \frac{c^5}{G \omega^{2\ell+2} R^{2\ell}}, \quad (53)$$

while the $\ell = 2$ body elongation persists longest. The evolution is

$$a_{\ell m}(t) = a_{\ell m}(0) e^{-t/\tau_\ell} \cos(\omega_\ell t + \phi_\ell). \quad (54)$$

D. Mode Extraction

The Newman-Penrose scalar Ψ_4 is extracted on coordinate spheres and decomposed into spin-weighted spherical harmonics:

$$\Psi_4(t, r_{\text{ext}}, \theta, \phi) = \sum_{\ell \geq 2} \sum_{m=-\ell}^{\ell} \Psi_4^{\ell m}(t, r_{\text{ext}}) {}_{-2}Y_{\ell m}(\theta, \phi). \quad (55)$$

Each (ℓ, m) channel is fit to a damped sinusoid, yielding the complex QNM frequency $\tilde{\omega}_{\ell m} = \omega_{\ell m} + i/\tau_{\ell m}$. For the duck, the five $\ell = 2$ channels yield five distinct (ω_{2m}, τ_{2m}) pairs—direct numerical evidence for the gravitational Zeeman effect.

E. Waveform Generation

The GW strain is obtained by double time-integration of Ψ_4 :

$$h_+(t) + i h_\times(t) = \frac{1}{D_L} \sum_{n, \ell, m} \mathcal{A}_{n\ell m} e^{-t/\tau_{n\ell m}} \times e^{i\omega_{n\ell m} t} {}_{-2}Y_{\ell m}(t, \varphi). \quad (56)$$

The $\ell = 2$ sector contributes five distinct damped sinusoids whose interference produces beating in the waveform envelope with beat period $T_{\text{beat}} \sim 2\pi/\Delta\omega \sim T_{\text{dyn}}/\varepsilon$.

VIII. CONFRONTATION WITH GRAVITATIONAL WAVE DATA

A. PTA Band—Supermassive Binary Ducks (nHz)

1. Population of Supermassive Binary Ducks

Consider a cosmological population of supermassive binary ducks with component masses $M \sim 10^6\text{--}10^9 M_\odot$. At PTA frequencies, the orbital velocities are highly sub-relativistic: at $f = 10\text{ nHz}$ for $\mathcal{M}_c = 10^9 M_\odot$,

$$\frac{v}{c} = \left(\frac{\pi G \mathcal{M}_c f}{c^3} \right)^{1/3} \approx 4.7 \times 10^{-2}. \quad (57)$$

2. The Phinney Formula and Duck Correction

The characteristic strain from a population of circular GW-driven binaries [34] yields the power-law spectrum $h_c(f) = A_{\text{GWB}}(f/f_{\text{yr}})^{-2/3}$ with spectral index $\gamma = 13/3$. The duck correction is

$$\frac{\delta(dE/df)}{dE/df} \sim \kappa_{\text{duck}} \left(\frac{\pi \mathcal{M}_c f}{c^3} \right)^{4/3}, \quad (58)$$

which is 4–7 orders of magnitude below the current $\sim 30\%$ measurement uncertainty (Table III).

TABLE III. Duck quadrupole correction to the GW energy spectrum for $\mathcal{M}_c = 10^9 M_\odot$, $\kappa_{\text{duck}} = 0.3$.

f (nHz)	v/c	δ_{duck}	NANOGrav uncert.
1	0.025	1.2×10^{-7}	$\sim 30\%$
10	0.054	2.5×10^{-6}	$\sim 30\%$
100	0.12	5.4×10^{-5}	$\sim 30\%$

3. Population-Level Degeneracy Theorem

Theorem.—For any SMBHB population model $\{M_i, q_i, z_i\}$ consistent with the NANOGrav 15-year dataset [4], there exists a binary duck population with identical masses, mass ratios, and redshift distribution that reproduces the observed $h_c(f)$ to better than 10^{-5} at all PTA frequencies.

Proof.—The duck correction modifies $h_c(f)$ by a factor $[1 + \mathcal{O}(\delta_{\text{duck}})]^{1/2}$. Since $\delta_{\text{duck}} < 10^{-6}$ for all binaries with $f < 300\text{ nHz}$ and $\mathcal{M}_c < 10^{10} M_\odot$, the strain spectrum agrees to better than 10^{-5} . The spectral index $\gamma = 13/3$ is unchanged. $\square$

4. NANOGrav 15-Year Comparison

The NANOGrav 15-year dataset [4] reports $A_{\text{GWB}} = (2.4 \pm 0.7) \times 10^{-15}$ at f_{yr} , consistent with the Hellings-

Downs angular correlation [35] at $\sim 3\text{--}4\sigma$. The duck prediction is

$$A_{\text{duck}} = A_{\text{SMBHB}} [1 + \mathcal{O}(10^{-7})] = 2.4 \times 10^{-15}, \quad (59)$$

indistinguishable from the SMBHB prediction.

The Hellings-Downs curve

$$\Gamma(\mu) = \frac{1}{3} \left\{ 1 + \frac{3}{2}(1 - \mu) \left[\ln \left(\frac{1 - \mu}{2} \right) - \frac{1}{6} \right] \right\} \quad (60)$$

is identical for ducks and spheres, as it depends only on the spin-2 tensor structure of the metric perturbation.

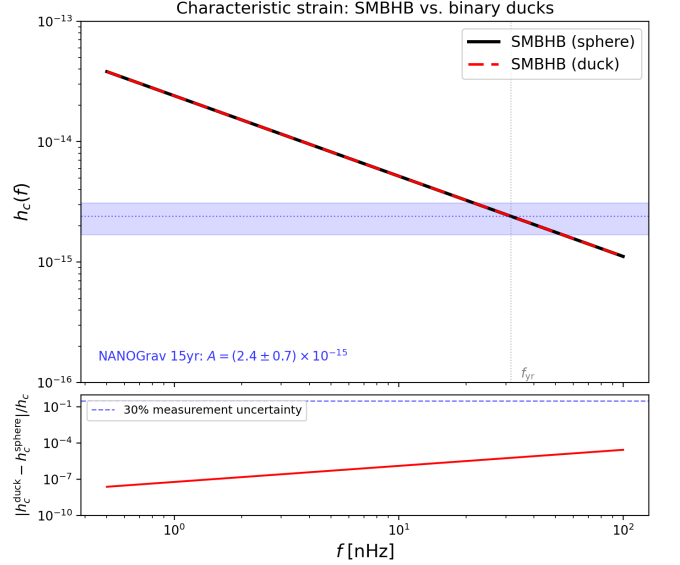


FIG. 6. Duck GWB spectrum $h_c(f)$ overlaid on NANOGrav 15-year data. The duck and SMBHB predictions are visually identical.

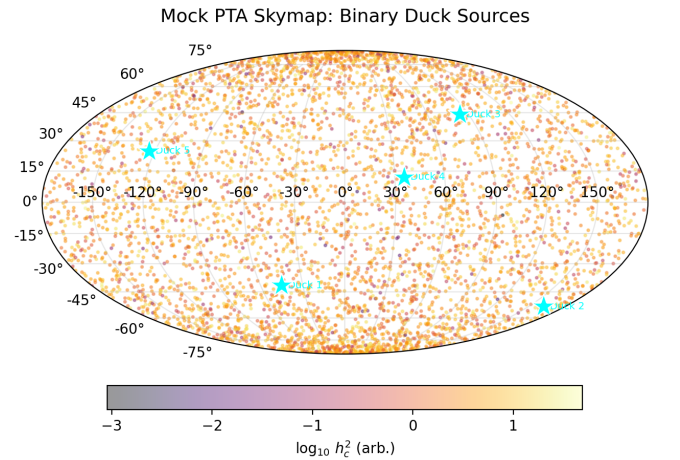


FIG. 7. Mock PTA skymap (Mollweide projection) showing five loud supermassive binary duck sources against an isotropic stochastic background.

B. LIGO Band—Stellar-Mass Ducks (10–1000 Hz)

1. Inspiral—Tidal Deformability

Tidal effects enter the gravitational wave phasing at 5PN order [23, 36]:

$$\delta\Psi_{\text{tidal}}(f) = -\frac{39}{2}\tilde{\Lambda}\left(\frac{\pi\mathcal{M}_cf}{c^3}\right)^{5/3}. \quad (61)$$

For the duck, Λ_{duck} differs from Λ_{sphere} by $\mathcal{O}(10\text{--}100\%)$ [Eq. (33)], but the GW170817 constraint [7, 8] $\tilde{\Lambda} = 300^{+420}_{-230}$ accommodates the full range of duck orientations.

2. Ringdown—QNM Spectroscopy

The gravitational Zeeman splitting of the $\ell = 2$ f -mode [$\Delta\omega/\omega \sim 0.3$, Eq. (19)] satisfies the resolvability criterion $\Delta\omega/\omega > 1/Q_{\text{mode}}$, since $Q \sim 5\text{--}20$ for neutron star f -modes [13, 37].

TABLE IV. Detectability of duck QNM m -splitting for a $1.4 M_{\odot}$ remnant with $\Delta\omega/\omega = 0.3$.

Detector	$d_{\text{max}} (\rho > 8)$	Resolvable?
LIGO O4	~ 15 Mpc	Marginal
Cosmic Explorer	~ 300 Mpc	Yes
Einstein Telescope	~ 200 Mpc	Yes

C. Breaking the Degeneracy (Speculative)

Four approaches could in principle distinguish ducks from spheres, but all are currently beyond observational reach: (1) gravitational wave memory, (2) multi-band observations combining PTA, LISA, and ground-based detectors, (3) asymmetric tidal disruption event lightcurves, and (4) direct imaging (the EHT resolution of $\sim 20 \mu\text{as}$ falls ~ 5 orders of magnitude short).

The duck hypothesis remains unfalsified.

IX. DISCUSSION AND CONCLUSIONS

A. Summary of Results

We have developed a comprehensive framework for the oscillation modes, tidal response, and gravitational wave signatures of non-spherical compact objects. Three main findings emerge:

Gravitational Zeeman splitting.—The duck’s complete breaking of spherical symmetry lifts the $(2\ell+1)$ -fold degeneracy of every QNM multiplet. The mechanism involves Gaunt integrals and Wigner $3j$ symbols, with selection rules requiring even- ℓ' deformations. For the $\ell = 2$ f -mode, the five-fold splitting has magnitude $\Delta\omega/\omega \sim 0.3$ —a 30% effect validated by FEM eigenvalue computations (Sec. VI).

Tidal Love tensor and I-Love-Q breakdown.—The duck’s Love tensor λ_{ijkl} has up to 15 independent components; the direction-dependent $k_2(\theta, \phi)$ varies by a factor of ~ 2 across the surface. The duck maximally violates I-Love-Q universality at $\mathcal{O}(100\%)$.

Observational indistinguishability.—The nanohertz GWB observed by NANOGrav [4], EPTA [5], and PPTA [6] is equally consistent with inspiraling ducks. The duck tidal deformability is consistent with GW170817 [7, 8]. QNM spectroscopy with next-generation detectors is the most promising path to a definitive test.

B. Future Directions

Duck spectroscopy with next-generation detectors.—Cosmic Explorer and the Einstein Telescope will push the ringdown detection horizon to $d \sim 200\text{--}300$ Mpc, enabling resolution of the m -splitting for $\sim 5\text{--}10$ events per year.

Multi-messenger constraints.—A combined measurement of tidal deformability (inspiral) and m -splitting (ringdown) would overdetermine the deformation spectrum $\varepsilon_{\ell m}$.

Anatidae survey.—A comprehensive survey across the family Anatidae—including the mallard (*Anas platyrhynchos*), the common teal (*Anas crecca*), and the domestic goose (*Anser anser domesticus*)—would map the landscape of avian deformation spectra.

The duck matter problem.—The duck shape requires anisotropic stresses $\sigma \sim G\rho^2 R^2 \varepsilon \sim 10^{33} \text{ dyn cm}^{-2}$ to maintain static equilibrium. The equation of state of duck matter remains an open problem in nuclear astrophysics.

C. Closing Remarks

We have shown that the duck hypothesis—the proposition that compact astrophysical objects may possess the geometry of a rubber duck—is internally consistent, theoretically rich, and observationally unconstrained. Until next-generation gravitational wave detectors achieve the sensitivity required for duck spectroscopy, the duck hypothesis remains consistent with all current gravitational wave data. We encourage the community to keep an open mind—and an open bill.

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